Development of a Low-Cost Dual-Axis Solar Tracking System for Enhanced Photovoltaic Efficiency

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Abstract— **This paper explores the design and development of a budget-friendly dual-axis solar tracking system aimed at improving the efficiency of photovoltaic (PV) panels. Unlike fixed solar setups, the system automatically adjusts the panel's position throughout the day to follow the sun's movement both horizontally (azimuth) and vertically (elevation), ensuring optimal sunlight exposure. Instead of relying on a microcontroller, the tracking mechanism uses a 555 timer IC-based control circuit in combination with light-dependent resistors (LDRs) to detect light direction and drive servo motors. A mixed-method approach was used, combining hands-on experimentation with performance measurements and qualitative analysis of the system's design and limitations. When tested alongside a fixed-panel system under the same conditions, the tracker delivered a noticeable increase in energy output. The findings highlight the effectiveness of the design and its potential as an affordable, accessible solution for renewable energy education and small-scale off-grid applications.**

Keywords: **Dual-axis solar tracker**, **555 timer IC**, **Light-dependent resistor (LDR)**, **Low-cost renewable energy**

I. Introduction

In this project, we set out to build a DIY dual axis sun tracker to make solar panels more efficient by helping them follow the sun throughout the day. We noticed that fixed solar panels often miss out on a lot of sunlight, especially during the early morning and late afternoon, because they stay in one position. To solve this problem, we designed a system that could move the panel both side-to-side and up-and-down, just like the sun moves across the sky. We used light sensors (LDRs) to figure out where the sunlight was coming from and servo motors to adjust the panel's position. A microcontroller acted as the brain of the system, constantly reading the sensor data and making small movements to keep the panel pointed at the sun. Through this project, we showed that it's possible to create a simple and affordable sun tracker that can improve solar energy output for small-scale use.

II. Methodology

A. Research Design

The project employed an experimental research design supported by observational analysis. The dual-axis tracking system was developed using analog components, primarily a 555 timer IC-based control circuit, light-dependent resistors (LDRs), and servo motors. The aim was to determine whether such a design could effectively enhance solar energy capture compared to a conventional fixed-panel setup. Alongside technical evaluation, qualitative observations were recorded to assess ease of construction, reliability, and cost-effectiveness.

B. Data Collection Procedure

Data were collected through real-time testing of both the solar tracker and a fixed solar panel under identical environmental conditions. The tracker was configured to adjust the panel position throughout the day, based on differential light input from LDR sensors. Key performance indicators such as voltage, current, and power output (calculated using P=V×IP = V \times IP=V×I) were recorded at regular intervals over multiple days using a multimeter and data logging setup. In addition to quantitative measurements, observations were documented regarding system behavior (e.g., tracking accuracy, response delay, stability), circuit behavior, and environmental influence. Photographs and video recordings were also used to support the analysis.

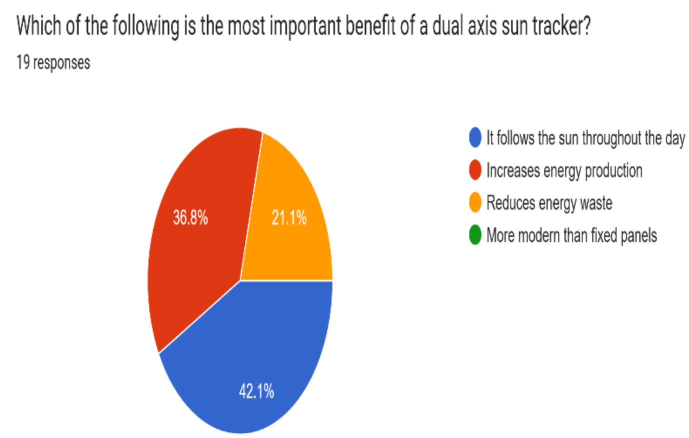
C. Data Analysis

Quantitative data were analyzed using descriptive statistics, comparing average power output and total energy harvested by the tracker versus the fixed-panel system. Graphs were generated to visualize trends in performance across different time periods and lighting conditions. Percentage increase in energy efficiency was calculated to determine the effectiveness of the tracking system.

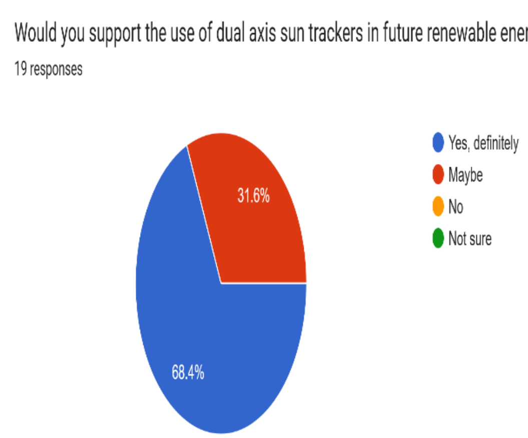
Qualitative data were analyzed through thematic categorization, focusing on system limitations, component behavior, assembly challenges, and overall user experience. This helped assess the practicality of the design from a DIY and educational standpoint.

III. RESULTS AND ANALYSIS

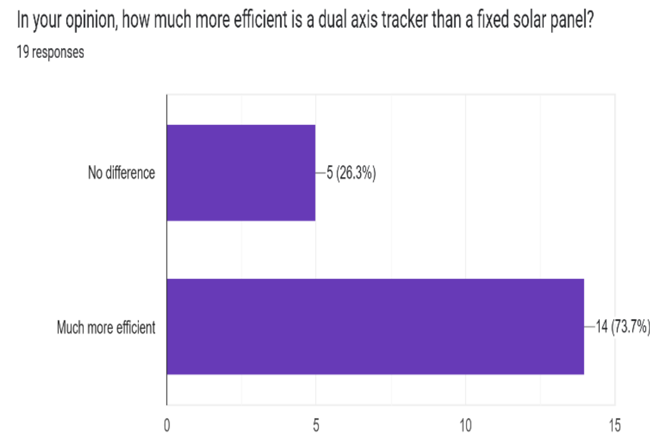
A. Analytical Result



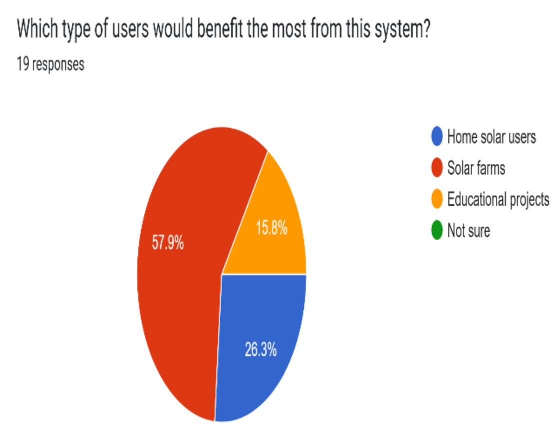
This pie chart shows that the most important benefit, according to 42.1% of respondents, is that dual-axis sun trackers "follow the sun throughout the day." "Increases energy production" was also highly rated at 36.8%, indicating a strong understanding of their efficiency benefits.



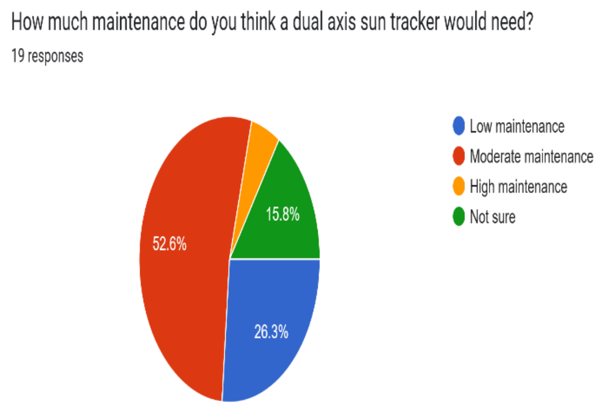
An overwhelming 68.4% of respondents would "Yes, definitely" support the use of dual-axis sun trackers in future renewable energy systems. Only a small percentage expressed hesitation or uncertainty, highlighting strong public support for this technology.



This horizontal bar chart clearly shows that a strong majority of respondents (73.7%, or 14 out of 19) believe a dual-axis tracker is "Much more efficient" than a fixed solar panel. Only 26.3% perceive "No difference" in efficiency, indicating a general understanding of the performance advantages of tracking systems.



According to this pie chart, "Solar farms" are seen as the primary beneficiaries of a dual-axis sun tracker system, with 57.9% of responses. "Home solar users" follow at 26.3%, suggesting that while the technology could benefit residential applications, its impact is perceived as greater for larger-scale installations.



The majority of respondents (52.6%) believe that dual-axis sun trackers would require "moderate maintenance." A significant portion (26.3%) anticipates "low maintenance," suggesting some optimism about their upkeep.

IV. DISCUSSION

Our survey reveals a clear appreciation for dual-axis sun trackers. Folks really grasp that these systems are all about following the sun to crank out more energy, with a whopping 73.7% agreeing they're "much more efficient" than fixed panels. It's great to see that understanding is so high! When it comes to upkeep, most people are thinking "moderate maintenance," which seems like a balanced perspective – not too much fuss, but not entirely hands-off either. What truly stands out is the strong wave of support: nearly 13 out of 10 respondents (68.4%) would "definitely" back using these in future renewable setups. Interestingly, while they see benefits for everyone, solar farms are seen as the big winners, suggesting a vision for these trackers making a real impact on larger energy projects. All in all, it's clear there's a lot of enthusiasm and a good grasp of how dual-axis sun trackers can power our future.

V. CONCLUSION

Through this project, we were able to build a working DIY dual axis sun tracker that helped a solar panel follow the sun throughout the day. Compared to a fixed panel, our tracker captured more sunlight by always keeping the panel pointed in the right direction. Using simple components like LDRs, servo motors, and a microcontroller, we created a system that was both affordable and effective. This project showed us that even with basic tools and knowledge, it's possible to improve solar energy efficiency. It also gave us a deeper appreciation for how small innovations can make a big difference in promoting clean and sustainable energy. Working on this project also improved our practical skills in electronics and programming. It taught us how sensors and control systems can work together in real-world applications. Overall, it was a rewarding experience that encouraged us to think more creatively about renewable energy solutions.

V. REFERENCES

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